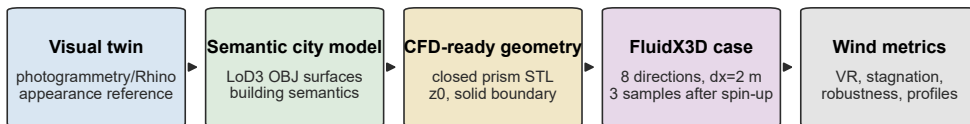
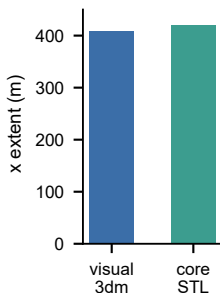


a

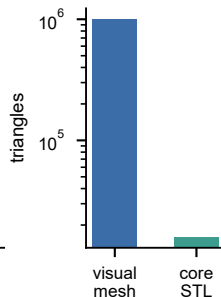


central claim: visual digital-twin assets define the scene, but CFD needs a semantic closed collision boundary

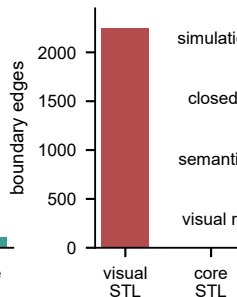
b matched study extent



c representation shift



d collision readiness



e geometry-to-CFD readiness logic

